Variation of Parameters

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

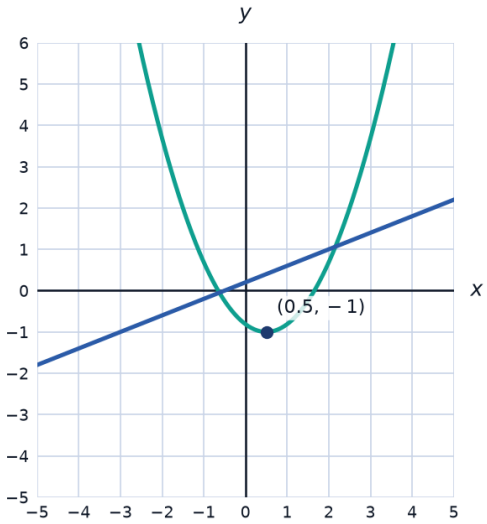
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: The Wronskian**

For $y_1 = e^x$, $y_2 = e^{2x}$ (solutions of $y'' - 3y' + 2y = 0$), compute the Wronskian $W = y_1y_2' - y_1'y_2$.

2 **[Foundational]** • **Target: The Wronskian**

For $y_1 = e^{3x}$, $y_2 = xe^{3x}$ (solutions of a repeated-root equation), compute W.



3 **[Foundational]** • **Target: The variation of parameters formula**

State the variation of parameters formula for a particular solution of $y'' + p(x)y' + q(x)y = g(x)$, given independent solutions y_1, y_2 of the homogeneous equation with Wronskian W.

4 **[Foundational]** • **Target: The variation of parameters formula**

Explain why the ODE must first be written in standard form $y'' + p(x)y' + q(x)y = g(x)$ (leading coefficient 1) before applying the variation of parameters formula.

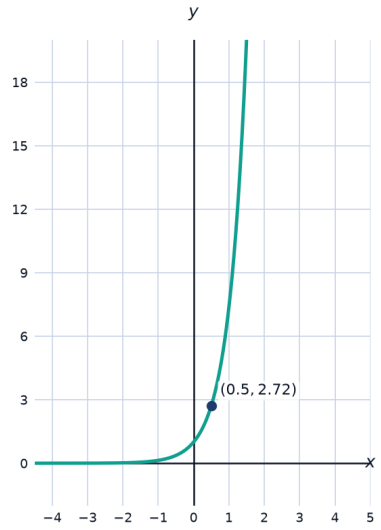
5 **[Foundational]** • **Target: A forcing term with no undetermined-coefficients trial form**

Solve $y'' + y = \tan x$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$ using variation of parameters.

© ClipSAT — clipsat.org | All Rights Reserved

6 [Foundational] • Target: Comparing with undetermined coefficients

Solve $y'' - y = e^{2x}$ by variation of parameters, then check the particular solution using undetermined coefficients.



7 [Foundational] • Target: Comparing with undetermined coefficients

Solve $y'' - y = 4$ (a constant forcing) by variation of parameters, and check against the obvious undetermined-coefficients guess $y_p = A$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: When to prefer each method

For $y'' + 4y = x^3 - 2x$, which method is more efficient: undetermined coefficients or variation of parameters? Explain.

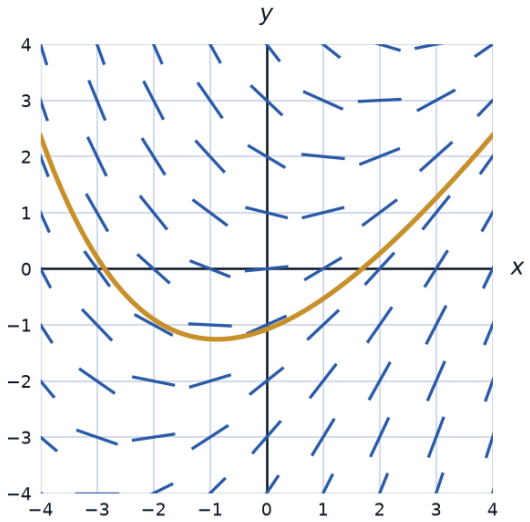
9 [Intermediate] • Target: When to prefer each method

For $y'' + y = \csc x$ (for $0 < x < \pi$), which method must be used, and why?

© ClipSAT — clipsat.org | All Rights Reserved

10 [Intermediate] • Target: Solving $y''+y=\csc x$ in full

Complete the solution of $y'' + y = \csc x$ $(0 < x < \pi)$ started above, using $y_1 = \cos x$, $y_2 = \sin x$, $W = 1$.



11 [Intermediate] • Target: Solving $y''+y=\csc x$ in full

Verify (by checking the coefficient of the leading term only) that $y_p = -x\cos x$ contributes a term of the correct order when substituted into $y'' + y$.

12 [Intermediate] • Target: Variation of parameters with a shifted independent variable

For the Cauchy–Euler-type substitution used in some ODE courses, $y_1 = x$, $y_2 = x^{-1}$ solve $x^2y'' + xy' - y = 0$ for $x > 0$. Rewrite this equation in the standard form $y'' + p(x)y' + q(x)y = g(x)$ needed for variation of parameters when the right side is $x^2 \cdot 3x = 3x^3$ (i.e., solve $x^2y'' + xy' - y = 3x^3$).

13 [Intermediate] • Target: Variation of parameters with a shifted independent variable

Continue the problem above: with $y_1 = x$, $y_2 = x^{-1}$, compute $W = y_1y_2' - y_1'y_2$.

14 [Intermediate] • Target: Variation of parameters with a shifted independent variable

Finish solving $x^2y'' + xy' - y = 3x^3$ $(x > 0)$ using $y_1 = x$, $y_2 = x^{-1}$, $W = -2x^{-1}$, and $g(x) = 3x$ (from the two questions above).

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

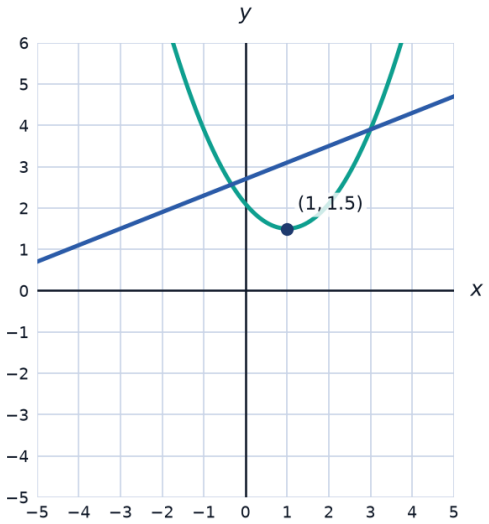
15 [Challenging] • Target: Variation of parameters with a shifted independent variable • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations

Check the particular solution $y_p = \frac{3}{8}x^3$ found above by substituting directly into $x^2y'' + xy' - y = 3x^3$.

© ClipSAT — clipsat.org | All Rights Reserved

16 [Challenging] • Target: A summary comparison • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

List two advantages of variation of parameters over undetermined coefficients.



17 [Challenging] • Target: A summary comparison • Cross-Lesson: 1.01 Separable Equations

List one advantage of undetermined coefficients over variation of parameters, when it applies.

18 [Challenging] • Target: A summary comparison • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

True or false, with a one-sentence justification: "Variation of parameters requires knowing the general solution of the associated homogeneous equation."

19 [Challenging] • Target: A summary comparison • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations

For $y'' + 4y = 8\tan 2x$ $(-\frac{\pi}{4} < x < \frac{\pi}{4})$, which method must be used, and what are y_1, y_2, and W?

20 [Challenging] • Target: A summary comparison • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

In one sentence, summarize when a student should reach for variation of parameters instead of undetermined coefficients.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

© ClipSAT — clipsat.org | All Rights Reserved

21 [Foundational] • Target: The Wronskian

Find a particular solution of $y'' - 6y' + 8y = 2e^{4x}$ (note: 4 is a root) using variation of parameters.

a. $xe^{4x} - \frac{e^{4x}}{2}$

b. $-xe^{4x} + \frac{e^{4x}}{2}$

c. $2xe^{4x} - e^{4x}$

d. $\frac{xe^{4x}}{2} - \frac{e^{4x}}{4}$

22 [Foundational] • Target: The variation of parameters formula

For $y'' + 4y = g(x)$, with $y_1 = \cos 2x$, $y_2 = \sin 2x$ ($W = 2$), which is the correct formula for $u_2'(x)$ in variation of parameters?

a. $\frac{g(x)\sin 2x}{2}$

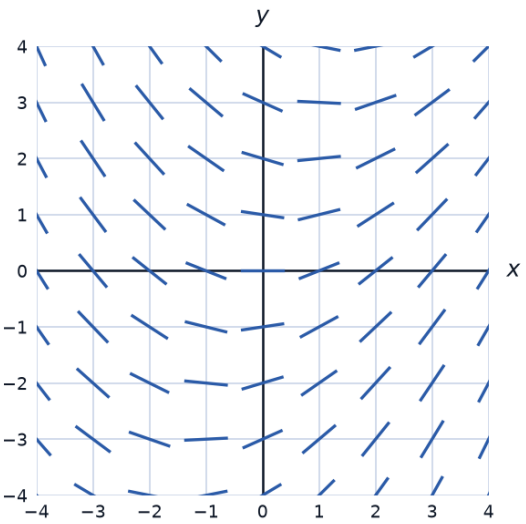
b. $\frac{g(x)\cos 2x}{2}$

c. $-\frac{g(x)\sin 2x}{2}$

d. $g(x)\cos 2x$

23 [Foundational] • Target: A forcing term with no undetermined-coefficients trial form

Consider the differential equation $y'' - xy = 0$. We are looking for a power series solution of the form $y(x) = \sum_{n=0}^{\infty} a_n x^n$ around the ordinary point $x = 0$. Find the coefficient a_2 in terms of a_0 and a_1.



a. $a_2 = a_0$

b. $a_2 = a_1$

c. $a_2 = 0$

d. $a_2 = a_0 + a_1$

© ClipSAT — clipsat.org | All Rights Reserved

24 [Foundational] • Target: A repeated root, with forcing dividing by x

Find a particular solution of $y'' + 16y = -\sec 4x$ (on an interval where $\sec 4x$ is continuous) using variation of parameters.

a. $\frac{x\sin(4x)}{4} + \frac{\log(\cos(4x))\cos(4x)}{16}$

b. $-\frac{x\sin(4x)}{4} - \frac{\log(\cos(4x))\cos(4x)}{16}$

c. $-\frac{x\sin(4x)}{2} - \frac{\log(\cos(4x))\cos(4x)}{8}$

d. $-\frac{x\sin(4x)}{8} - \frac{\log(\cos(4x))\cos(4x)}{32}$

25 [Foundational] • Target: Comparing with undetermined coefficients

For the classical RK4 method, dividing the step size h by 10 reduces the global error by roughly what factor?

a. 10

b. 10000

c. 100

d. 20

26 [Foundational] • Target: When to prefer each method

For Euler's method, dividing the step size h by 10 reduces the global error by roughly what factor?

a. 10

b. 100

c. 10000

d. 20

27 [Foundational] • Target: Solving $y''+y=\csc x$ in full

For the improved Euler (Heun) method, dividing the step size h by 3 reduces the global error by roughly what factor?

a. 3

b. 9

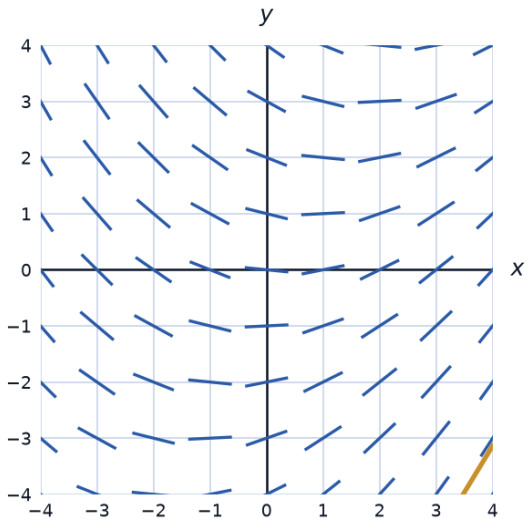
c. 27

d. 81

© ClipSAT — clipsat.org | All Rights Reserved

28 [Foundational] • Target: Variation of parameters with a shifted independent variable

Consider the differential equation $\frac{dy}{dx} = x + y$ with the initial condition $y(0) = 1$. Use the Euler method with a step size of $h = 0.1$ to approximate the value of $y(0.2)$.



a. 1.20

b. 1.21

c. 1.22

d. 1.23

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A summary comparison

Consider the system of linear differential equations given by: $\frac{dx}{dt}, = x + 2y$; $\frac{dy}{dt}, = 3x + 2y$ If the solution to this system is of the form $x(t) = c_1e^{\lambda_1 t} + c_2e^{\lambda_2 t}$ and $y(t) = k_1c_1e^{\lambda_1 t} + k_2c_2e^{\lambda_2 t}$, what is the value of the larger eigenvalue λ?

a. 6

b. 5

c. 4

d. 3

30 [Intermediate] • Target: The Wronskian

Consider the differential equation $\frac{dy}{dx} = x + y$ with the initial condition $y(0) = 1$. Use the Euler method with a step size of $h = 0.1$ to approximate $y(0.2)$.

a. 1.200

b. 1.210

c. 1.230

d. 1.220

© ClipSAT — clipsat.org | All Rights Reserved

31 **[Intermediate]** • **Target: The variation of parameters formula**

Consider the initial value problem $y'' + 4y = \delta(t - \pi)$ with $y(0) = 0$ and $y'(0) = 0$. Which of the following represents the Laplace Transform of the solution, $Y(s) = \mathcal{L}\{y(t)\}$?

a. $\frac{e^{-\pi s}}{s^2 + 4}$

b. $\frac{1}{s^2 + 4}$

c. $\frac{e^{-\pi s}}{s^2}$

d. $\frac{e^{-\pi s}}{s^2 - 4}$

32 **[Intermediate]** • **Target: A forcing term with no undetermined-coefficients trial form**

Solve $\frac{dy}{dx} = \frac{y \ln x}{x(y^2 + 1)}$ for $x > 0, y > 0$, with $y(1) = 1$. What is the numerical value of $y^2 + 2 \ln y$ when $x = e^2$?

a. 5

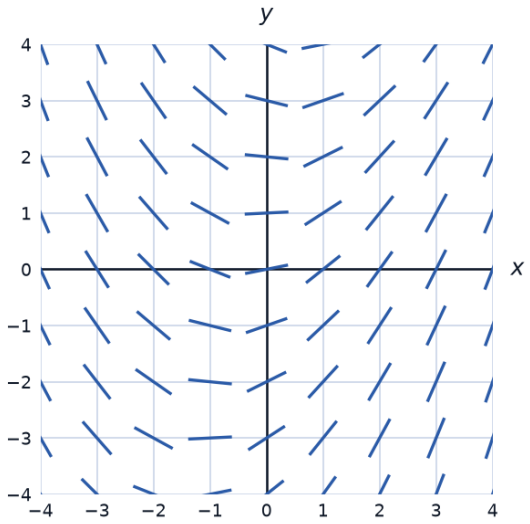
b. 6

c. 4

d. 3

33 **[Intermediate]** • **Target: A repeated root, with forcing dividing by x**

Solve $y' + 2y = 4t$ with $y(0) = y_0$. For what value of y_0 does the solution contain no exponential transient term?



a. 0

b. -1

c. 1

d. $-\frac{1}{2}$

© ClipSAT — clipsat.org | All Rights Reserved

34 [Intermediate] • Target: Comparing with undetermined coefficients

Solve $\frac{dy}{dx} = \frac{x^2+2}{y^2-1}$ with $y(0) = 2$ for $y > 1$. What is the value of $\frac{1}{3}y^3 - y$ when $x = 3$?

a. 15

b. $\frac{49}{3}$

c. $\frac{47}{3}$

d. $\frac{45}{3}$

35 [Intermediate] • Target: When to prefer each method

Given $\frac{dy}{dx} = \frac{x^2+1}{y^2+1}$ with $y(0) = 0$, what is the numerical value of $\frac{1}{3}y^3 + y$ when $x = 3$?

a. 12

b. 15

c. 10

d. 9

36 [Intermediate] • Target: Solving $y''+y=\csc x$ in full

Which recurrence do the coefficients of $y = \sum a_n x^n$ satisfy for $y'' - xy' - y = 0$?

a. $a_{n+2} = \frac{a_n}{(n+2)(n+1)}$

b. $a_{n+2} = \frac{a_n}{n+2}$

c. $a_{n+2} = -\frac{a_n}{n+2}$

d. $a_{n+2} = \frac{(n+1)a_n}{n+2} \cdot 2$

37 [Intermediate] • Target: Variation of parameters with a shifted independent variable

Use the improved Euler (Heun) method with $h = \frac{1}{2}$ to approximate $y(\frac{1}{2})$ for $y' = y - x$, $y(0) = 2$.

a. $\frac{25}{8}$

b. 3

c. 2

d. $\frac{9}{2}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

© ClipSAT — clipsat.org | All Rights Reserved

38 [Challenging] • Target: A summary comparison • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

On $[0, 1]$, Euler's method with $h = 0.1$ gives an error of about $\frac{1}{50}$. Roughly how many steps are needed for an error of $\frac{1}{500}$?

a. 100

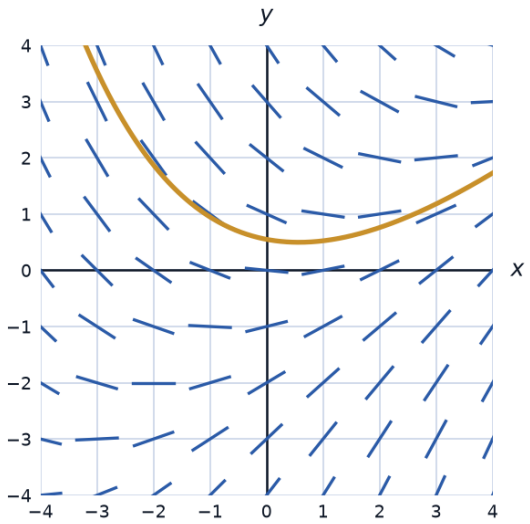
b. 1000

c. 200

d. 50

39 [Challenging] • Target: The Wronskian • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations

On $[0, 1]$, the improved Euler method with $h = 0.1$ gives an error of about $\frac{1}{50}$. Roughly how many steps are needed for an error of $\frac{1}{200}$?



a. 40

b. 11

c. 30

d. 20

40 [Challenging] • Target: The variation of parameters formula • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

Use the classical RK4 method with $h = \frac{1}{2}$ to approximate $y(\frac{3}{2})$ for $y' = x - y$, $y(1) = 2$.

a. $\frac{3}{2}$

b. $\frac{329}{192}$

c. 2

d. $\frac{7}{4}$

© ClipSAT — clipsat.org | All Rights Reserved

41 [Challenging] • Target: A forcing term with no undetermined-coefficients trial form • Cross-Lesson: 1.01 Separable Equations

For $y' = -y$, $y(0) = 1$, Euler's method with $h = \frac{1}{4}$ gives $y(1) \approx \frac{81}{256}$. What is the actual error $y(1) - y_4$?

a. $\frac{81}{256} - e^{-1}$

b. $-\frac{81}{256} + e^{-1}$

c. $\frac{81}{256}$

d. e^{-1}

42 [Challenging] • Target: A repeated root, with forcing dividing by x • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

What is the global truncation error of the improved Euler (Heun) method?

a. $O(h^2)$

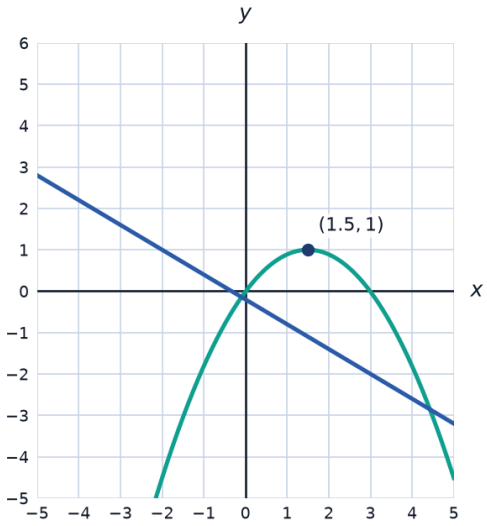
b. $O(h)$

c. $O(h^3)$

d. $O(h^4)$

43 [Challenging] • Target: Comparing with undetermined coefficients • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations

What is the global truncation error of the classical RK4 method?



a. $O(h)$

b. $O(h^4)$

c. $O(h^2)$

d. $O(h^3)$

© ClipSAT — clipsat.org | All Rights Reserved

44 [Challenging] • Target: When to prefer each method • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

Use Euler's method with $h = \frac{1}{4}$ to approximate $y(\frac{1}{4})$ for $y' = -y$, $y(0) = 3$.

a. $\frac{75}{32}$

b. 3

c. $\frac{3}{2}$

d. $\frac{9}{4}$

45 [Challenging] • Target: Solving $y''+y=\csc x$ in full • Cross-Lesson: 1.01 Separable Equations

What is the global truncation error of Euler's method?

a. $O(h^2)$

b. $O(h^3)$

c. $O(h^4)$

d. $O(h)$

© ClipSAT — clipsat.org | All Rights Reserved